Advanced Software Engineering (IT561)

Technical Elective Course (3-0-2-4)

BTech Sem-VI / Sem-VIII (ICT+CS), MTech Sem-II (ICT-SS)

Prerequisites

- Students should have completed a course in fundamentals on software engineering and programming, familiarity with data structures and basic algorithms, and exposure to at least one mainstream programming language and version control.

Course Objectives

Goal of the course is to provide students with a reasonable understanding of selected advanced topics in software engineering that are relevant to the changing industry demands. Key objectives have been summarized below.

- Architect medium-to-large systems and justify tradeoffs quantitatively.
- Specify and verify nontrivial functional and nonfunctional requirements.
- Design testable systems and produce comprehensive test suites plus CI/CD automation.
- Build resilient, observable distributed services and demonstrate incident handling.
- Evaluate alternatives using decision analysis and empirical metrics, and ethically justify design choices.

Key Topics

- SE Foundations: What makes SE hard; lifecycle overview.
- Requirements engineering & traceability, QAs/Non-functional reqs.
- Software architecture: styles, documentation, ATAM basics.
- Design patterns, architectural tactics & refactoring.
- Verification, static/dynamic analysis & program analysis.
- Testing strategies at scale.
- DevOps, CI/CD, build and release engineering.
- Observability, monitoring, and incident response.
- Resilience engineering and chaos testing
- Microservices, APIs and service meshes
- Data-intensive systems: consistency, streaming, and storage choices
- Maintenance, product lines, ethics & regulatory compliance, Technical debt.

Grading Components

- Class test, In-sem & End-sem exam, lab assignments, group project, research paper reading.

Evaluation Scheme

- **Lab(50%)**
  - Assignments & Labs: 20%
  - Team Project: 20%
  - Research paper reading/presentation: 10%
- **Theory(50%)**
  - Class Tests: 10%
  - In-Sem Exam: 20%
  - End-Sem Exam: 20%

Course Outcomes

- Understanding modern software engineering paradigms.
- Develop conceptual understanding in architectural decision-making using evaluative models.
- Getting with configuring CI/CD pipelines, implementing infrastructure as code, and integrating observability components (logging, tracing, monitoring) for contemporary systems.
- Understanding to apply techniques in scalability, distributed consensus, and security for development of enterprise-grade systems.
- Understanding basics of software evolution and refactoring.
- Developing research-oriented thinking in software engineering.

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X	X	X			X	X			